Robustness is not Reliability: Silent Conformal Coverage Failure under Operational Spectral and Domain Shift in Sentinel-2 Cloud Segmentation

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Abstract

Operational Earth observation increasingly relies on deep segmentation models whose predictions feed automated pipelines, and a growing body of work measures how well these models retain their *accuracy* under the distribution shifts that operational deployment entails: atmospheric-correction processing, sensor changes, unseen geographies and surfaces. We show that accuracy-robustness is not the property that matters for trustworthy deployment, and that a conformal certificate calibrated before deployment can be actively misleading. Using split conformal risk control on real Sentinel-2 cloud segmentation (CloudSEN12), we demonstrate that under the operational L1C→L2A atmospheric-correction shift a model's *reliability degrades far faster than its accuracy, and silently*: pixel accuracy falls a moderate 12 points ($79.6 \rightarrow 67.6\,\%$) while a clean-calibrated conformal threshold's confidently-wrong (joint) risk rises more than threefold, from $8.4\,\%$ to $28.9 \pm 0.3\,\%$ against a $10\,\%$ target—a $+20$-point rise the calibration statistic it minimises never reports, because that statistic stays below target by construction. Per-stratum (Mondrian) recalibration restores the guarantee ($8.7 \pm 0.2\,\%$; paired gap $+20.2 \pm 0.4$, two-way cluster-robust standard error over 30 runs). We then show the failure is diagnosable rather than universal. Among pure deployment shifts (input held fixed), a structural surface gap (deploying on unseen bright surfaces) breaks the naive certificate ($11.4 \pm 0.3\,\%$, its confidence interval excluding the target, restored to $2.5\,\%$), while a milder geographic gap (train Northern hemisphere, deploy Southern) does not ($9.5\,\%$, just below the target). The reliability crisis therefore requires a *structural* domain gap, not any shift. It is also model-class-dependent: a channel-adaptive foundation model (DOFA), run through the *identical* protocol, is far more reliable under the atmospheric shift (naive joint risk $9.9\,\%$ versus $28.9\,\%$ for the small model, sitting at the target rather than far above it), though it too breaks under a spectral-band-drop shift. Throughout, we report every risk with its coverage, because the certified quantity is a joint mass that abstention alone can drive to zero. Methodologically, a defensible certificate on this data requires a scene-connected-component exchangeable unit (Sentinel-2 products that span multiple annotation regions violate ROI-level exchangeability) and a two-way cluster-robust standard error over the crossed calibration-split $\times$ model-seed design. To our knowledge this is the first operational-remote-sensing instantiation of the robustness $\neq$ reliability decoupling, together with a diagnosis of *when* it occurs, on real Sentinel-2.

Keywords: conformal prediction, uncertainty quantification, distribution shift, cloud segmentation, Sentinel-2, operational reliability

1. Introduction

Earth-observation segmentation models are moving from research benchmarks into operational pipelines: cloud masking that gates every downstream surface-reflectance product, land-cover mapping that feeds policy, change detection that triggers alerts. In these settings the model rarely sees the exact distribution it was trained on. The Sentinel-2 archive alone exposes a model to at least four operational shifts: the L1C→L2A atmospheric correction that converts

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top-of-atmosphere radiance to bottom-of-atmosphere reflectance and removes the cirrus band B10; cross-sensor transfer as constellations evolve; deployment across geographies and seasons absent from the training set; and deployment on surface types (bright built-up, bare soil, snow and ice) that are both operationally critical for cloud confusion and under-represented in labelled data.

A substantial literature now measures how well models *maintain accuracy* under such shifts. Channel- and sensor-adaptive encoders [21, 20, 6, 3] are evaluated on their cross-band and cross-sensor accuracy; robustness benchmarks score accuracy under image-space corruptions [9]; cross-band generalisation studies quantify accuracy transfer across spectral-band configurations [16]. What this literature does not ask is whether, *under the same shifts*, the model still **knows when it is wrong**—whether the uncertainty it reports remains trustworthy. That question is not a refinement of accuracy-robustness; the two can decouple. A model can preserve its mean accuracy while its confidence-to-error relationship silently drifts, so that a fixed operating point calibrated before deployment admits far more confident errors than it certifies.

Conformal prediction offers a principled way to *certify* such an operating point: split conformal and its risk-controlling variants give a finite-sample guarantee on a chosen error notion. But the guarantee holds only under exchangeability between calibration and test data, and operational shift is precisely a controlled violation of exchangeability. The danger is not that the certificate fails loudly—it is that it fails *silently*: the calibration-set statistic the procedure minimises stays below target by construction, so a practitioner reading only that number sees success, while the deployed risk runs above target with nothing to announce it.

This paper makes that failure concrete, operational, and—crucially—bounded. Our contributions are:

- **C1 — Reliability degrades faster than accuracy, and silently, under a real atmospheric shift.** On CloudSEN12, under the genuine L1C→L2A Sen2Cor transition (not a synthetic ablation), pixel accuracy falls a moderate 12 points while a clean-calibrated conformal threshold's confidently-wrong joint risk more than triples to $28.9\,\%$ against a $10\,\%$ target. The reliability breach is disproportionate to the accuracy loss and is never announced, since the certificate's own calibration statistic still passes (Section 4.2).
- **C2 — Whether the failure occurs is diagnosable: it needs a structural domain gap, and its severity is model-class-dependent.** Two pure deployment shifts (input held fixed) separate the conditions: a structural surface gap breaks the naive certificate ($11.4\,\%$), while a milder geographic gap does not ($9.5\,\%$, just below target). Together these give a diagnosis of *when* naive conformal fails—under a structural domain gap, not under any shift (Sections 4.3–4.4).
- **C3 — A principled, honestly-costed remedy.** Per-stratum (Mondrian) recalibration restores control on both completed axes; because the certified quantity is a joint mass that abstention alone can drive to zero, we report every risk with its coverage and quantify the coverage the remedy spends.
- **C4 — The methodology a defensible certificate on this data requires.** The exchangeable unit must be the scene-connected component, not the annotation ROI—12 Sentinel-2 products in the CloudSEN12 test split each appear under two ROIs, so ROI-level splitting leaks a scene across calibration and test—and the standard error over the crossed calibration-split $\times$ model-seed design must be two-way cluster-robust rather than treating the 30 cells as independent (Section 3).

We are deliberate about scope. We claim only the *operational instantiation*: that naive conformal coverage can fail under distribution shift is established theory (the coverage-gap bound of Barber et al. [4], and the online treatment of Gibbs and Candès [7]), with weighted conformal a known remedy under known covariate shift [18]; the risk-control machinery is borrowed [1]; and per-stratum recalibration is a Mondrian special case [19]. Our mechanism components (band-group tokenisation, wavelength-conditioned attention, spectral-group masked pretraining, band-group dropout) are cited reused ingredients, not claimed contributions. What is new is the demonstration, diagnosis and honest costing of this failure for operational remote-sensing segmentation on real Sentinel-2, and the exchangeability/variance methodology that makes the demonstration defensible.

2. Related work

Conformal prediction under distribution shift.. Split conformal prediction gives distribution-free finite-sample coverage under exchangeability; when calibration and test distributions differ, coverage is no longer guaranteed. Weighted conformal prediction [18] recovers validity under *known* covariate shift; Barber et al. [4] bound the coverage gap under general non-exchangeability; Gibbs and Candès [7] and Podkopaev and Ramdas [14] treat online and label shift. Conformal risk control [1] generalises coverage to monotone risks and is the machinery we adopt. We use none of these to *fix* the shift—our naive arm is deliberately an unweighted certificate, so that its failure is the measurement; the remedy we study is the simplest valid instance, per-stratum (Mondrian, Vovk et al. 19) recalibration on discrete operational states.

Conformal and selective prediction in remote sensing.. Conformal methods have reached remote sensing but almost exclusively *in distribution*: split-conformal classification on hyperspectral benchmarks with validity demonstrated but no shift and no stratification [SACP, 10]. Our work differs in testing the certificate *under operational shift* and in isolating the exchangeable unit that makes the certificate defensible on real-scene data.

Robustness $\neq$ reliability.. The decoupling of accuracy-robustness from uncertainty-reliability has been observed outside remote sensing—under natural and medical distribution shift, chiefly through calibration and expected-calibration-error rather than conformal risk [15, 12]. We give what is, to our knowledge, the first remote-sensing, conformal, operational-shift instantiation.

Channel-adaptive and missing-band remote sensing.. Sensor- and channel-adaptive encoders [21, 20, 6] and channel-sampling regularisers [3, 13] target *accuracy* under varying or missing bands. We engage this literature as the source of our (cited) mechanism components and, in a supporting decomposition (Section 4.5), by showing which ingredients actually carry missing-band robustness and how that carriage is itself regime-dependent.

3. Method

3.1. Task, data, and models

We study per-pixel cloud segmentation on CloudSEN12 [2], a Sentinel-2 benchmark of expert-labelled cloud and cloud-shadow masks with co-registered L1C (top-of-atmosphere, 13 bands) and L2A (Sen2Cor bottom-of-atmosphere, 12 bands; B10 removed, Main-Knorn et al. 11) products for the identical pixels. We deliberately study per-pixel spectral segmentation without spatial context, so that the reliability signal we measure is spectral and not a function of spatial redundancy; a spatial-context foundation-model baseline is included in Section 4.4. The classes are clear, thick cloud, thin cloud, and cloud shadow.

Our primary model is a band-as-modality encoder: physically-grouped spectral bands are tokenised, given a wavelength-conditioned positional encoding, mixed by cross-band attention, and pretrained with a spectral-group masked autoencoder before supervised fine-tuning with band-group dropout. This architecture is not a contribution—every ingredient is a cited reused component—but a substrate on which to measure reliability; we include a plain multilayer-perceptron baseline (with and without band-group dropout) and, in Section 4.4, a frozen channel-adaptive foundation model [DOFA, 21].

3.2. Reliability framework

For a fixed trained model we score every test pixel by its maximum softmax probability after temperature scaling, and study the operating point selected by split conformal *risk* control (CRC) at a target risk $\alpha = 10\,\%$. The certified quantity is the *joint* confidently-wrong mass $P(\text{accepted} \wedge \text{wrong})$: the fraction of pixels the model both accepts (confidence above threshold) and misclassifies. CRC selects the lowest threshold whose calibration-set upper bound on this mass is $\leq \alpha$; the guarantee is a marginal expectation over the joint draw of calibration and test, under exchangeability of the per-unit loss.

We contrast two arms at each shift state. The **naive** arm calibrates the threshold and temperature once, on the source (clean / in-training) distribution, and deploys them under shift. The **Mondrian** arm recalibrates the threshold and temperature on the shifted (target) state itself. A third **control** arm (source threshold, target-state temperature) attributes any naive failure to the stale threshold versus stale temperature. Because a joint mass is driven to zero by abstaining, we report every risk together with its coverage (the fraction of pixels accepted); a risk reduction bought only by accepting fewer pixels is reported as such, not as a win.

Table 1: Flagship reliability under the atmospheric shift (proposed model, 30 runs, two-way cluster-robust SE). All values in %.

state	naive joint risk	control	Mondrian	naive cov.	Mondrian cov.
clean (L1C)	8.45	8.45	8.45	76.3	76.3
L2A (real)	$\mathbf{28.93 \pm 0.34}$	29.80 ± 0.57	$\mathbf{8.72 \pm 0.24}$	93.7	47.4

3.3. Exchangeable unit and valid uncertainty

Two methodological choices are load-bearing for a defensible certificate on real-scene data, and we found both by adversarial review of an earlier version of this study.

Exchangeable unit. CRC's guarantee is over exchangeable per-unit losses. The obvious unit—the annotation ROI—is not exchangeable here: CloudSEN12 ships several dated patches per ROI, and, more subtly, 12 Sentinel-2 products in the test split each appear under two different ROIs, so two ROIs treated as independent can be the same acquisition, tile and atmosphere. We therefore union ROIs that share any Sentinel-2 product identifier into *scene-connected components*. This merges 11 units (195 test ROIs become 184 components; the count is 11, not 12, because two of the shared products chain through a common ROI into a single three-ROI component), and we use the component as the exchangeable unit on both the three-way calibration/temperature/evaluation split and the CRC grouping.

Valid standard error. Our estimand is the mean held-out risk over a crossed design of 10 calibration splits $\times$ 3 model seeds. The 30 cells are not independent: a model seed recurs across all 10 splits, and a split recurs across all 3 seeds. A plain standard-error-of-the-mean therefore understates uncertainty. We report a two-way cluster-robust standard error [5] that accounts for both clustering directions and degrades to the independent estimate when neither factor clusters. Temperature scaling is fitted on a *disjoint* set of scene-components from the CRC calibration set: because multi-class maximum-softmax ordering is not temperature-invariant, fitting the temperature on the calibration pixels would break the exchangeability the theorem needs.

4. Experiments

4.1. Setup

CloudSEN12, real Sentinel-2, 195 test ROIs $\rightarrow$ 184 scene-connected components, target risk $\alpha = 10\,\%$, 30 runs (10 calibration splits $\times$ 3 model seeds), two-way cluster-robust standard errors for the flagship (Section 4.2), deployment-axis (Section 4.3) and DOFA-baseline (Section 4.4) conformal experiments, all on the scene-connected-component unit; the physical-degradation decomposition (Section 4.5) uses its own design and error model, stated in place.

4.2. Flagship: a silent reliability crisis under the atmospheric shift

Under the operational L1C$\rightarrow$L2A shift the naive certificate fails and Mondrian recalibration restores it, and the reliability breach is disproportionate to the accuracy loss (Table 1, Fig. 1).

Three readings. First, the disproportion is the point: across the same shift, pixel accuracy falls a moderate 12 points ($79.6\,\%$ clean $\rightarrow$ $67.6\,\%$ on L2A) while the naive joint risk more than triples ($8.4 \rightarrow 28.9\,\%$), a $+20.2 \pm 0.4$ paired gap versus Mondrian—reliability degrades far faster than accuracy, and silently, since the calibration statistic the procedure minimises stays $\leq \alpha$ throughout. An operator watching accuracy would see a tolerable degradation; the uncertainty they rely on to abstain has by then failed. Second, the control arm (fresh temperature, stale threshold) is no better than naive—in fact marginally worse (29.8 versus $28.9\,\%$)—so the breach is attributable to the stale *threshold*, not to temperature miscalibration: recalibrating confidence alone would not save a deployed detector. Third, and essential to honesty, Mondrian restores validity partly by accepting far fewer pixels (coverage $47\,\%$ versus $94\,\%$): the certified quantity is a joint mass, and part of the remedy is principled abstention. We report this cost rather than quoting the risk alone.

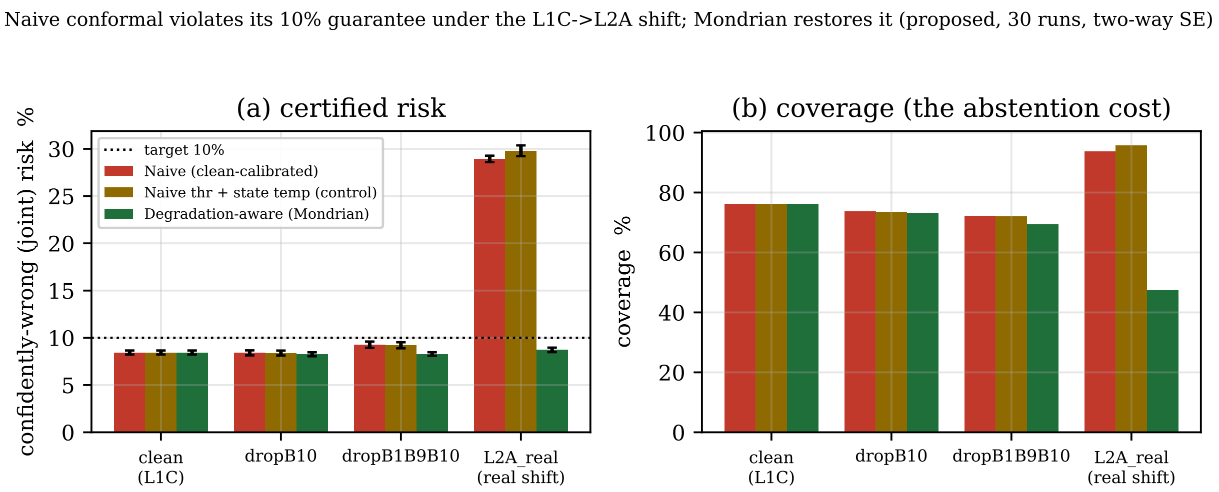

Figure 1: Naive conformal violates its $10\,\%$ guarantee under the L1C→L2A shift while Mondrian restores it (proposed model, 30 runs, two-way cluster-robust SE). (a) certified joint risk per state; (b) coverage, the abstention cost of the remedy.

4.3. Two pure deployment shifts: a surface gap that breaks, a geographic gap that does not

To test whether the failure is specific to the atmospheric shift we construct two further shifts of a different *type* on the same task, holding the input product fixed (clean L1C) so that the shift is *where the model is deployed*, not *what bands it sees*. In each, the naive arm calibrates on held-out source units and is evaluated on unseen target units; the Mondrian arm recalibrates on the target (Fig. 2).

Surface domain gap. Training and naive-calibrating only on dark surfaces (vegetation, cropland, water) and deploying on the bright surfaces (built-up, bare soil, snow/ice) the model has never seen—the operational cloud-confusion case—breaks the naive certificate: joint risk $11.38 \pm 0.34\,\%$, its confidence interval $[10.7, 12.1]$ excluding the $10\,\%$ target, restored by Mondrian to $2.47 \pm 0.17\,\%$. The plain-MLP-with-band-group-dropout baseline (b2) mirrors this ($11.9 \rightarrow 2.5\,\%$), so the effect is not architecture-specific.

Geographic domain gap. Training and calibrating on the Northern hemisphere and deploying on the unseen Southern hemisphere, under the identical 30-run design, does *not* break the certificate: the proposed model's naive joint risk on Southern units is $9.52 \pm 0.24\,\%$ ($95\,\%$ CI $[9.05, 9.98]$, just below the $10\,\%$ target), and the plain-MLP baseline straddles it ($9.82 \pm 0.22\,\%$, CI $[9.4, 10.3]$). Neither is the clear violation the surface gap produced. Accuracy also transfers well ($79\,\%$ on Southern units), consistent with Northern and Southern mid-latitudes sharing surface and cloud regimes. Mondrian still lowers the risk further (proposed $5.9\,\%$), but from an already-controlled starting point.

The surface and geographic axes are the paper's cleanest comparison, because both are pure deployment shifts on a fixed input and so lie on one scale—they differ only in the structural size of the domain gap. The structural surface gap (dark training surfaces to spectrally very different bright ones) breaks the naive certificate; the milder geographic gap (regions that share surface and cloud regimes) does not. This is direct evidence that the reliability failure requires a *structural* domain gap, not mere geographic distance or any shift, and it is what lets us report a diagnosis of *when* naive conformal fails operationally rather than a claim that it always does. The atmospheric shift breaks harder still ($28.9\,\%$) but also changes the input representation, so we read it as a stronger, different-mechanism confirmation rather than another point on the same deployment-gap scale.

On the surface axis the Mondrian arm recalibrates on a median of only $\sim$12 scene-component units (versus $\sim$91 for the naive arm and $\sim$69 for the flagship). At $\alpha = 10\,\%$ this is near the conformal feasibility floor, so the restored $2.5\,\%$ at $41\,\%$ coverage reflects heavy, conservative over-abstention on a small target stratum rather than a tight guarantee at target—we report it as restoring *control at a large coverage cost*.

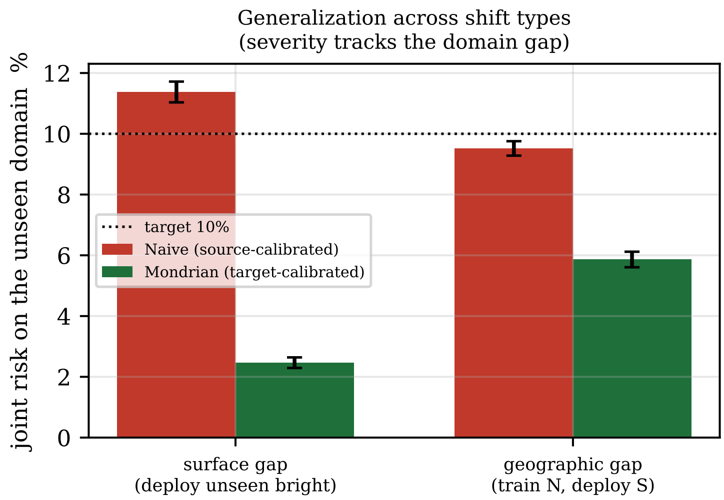

Figure 2: Naive versus Mondrian joint risk on the unseen target domain, for the surface and geographic deployment shifts (proposed model). The structural surface gap breaks the $10\,\%$ certificate; the milder geographic gap does not.

4.4. Model-class dependence

We run a frozen channel-adaptive foundation model [DOFA, 21] through the *identical* flagship protocol—the same CRC joint-risk estimand, the scene-connected-component unit, the 10×3 crossed design, and the two-way cluster-robust SE—so the comparison is like-for-like. DOFA is *far more* reliable than the small per-pixel model under the L1C→L2A shift: naive joint risk $9.87 \pm 0.17\,\%$ versus $28.9\,\%$ for the small model. Its L2A joint risk sits right at the $10\,\%$ target ($95\,\%$ CI $[9.53, 10.20]$, straddling it), so a clear residual breach cannot be established; Mondrian lowers it to $8.73 \pm 0.18\,\%$, and the abstention cost is far gentler than the flagship's (coverage $79.7 \rightarrow 76.5\,\%$, versus $94 \rightarrow 47\,\%$ for the small model). DOFA is not immune, however: under a spectral-band-drop shift (removing its two SWIR bands) its naive joint risk does breach the target ($11.56 \pm 0.55\,\%$, CI $[10.49, 12.63]$ excluding 10), restored by Mondrian to $8.55\,\%$. The reliability crisis is thus strongly *model-class-dependent*: a channel-adaptive foundation model largely escapes it under the atmospheric shift but still breaks under a band-drop shift, and we do not claim a universal failure. (This is a foundation-model *feature* baseline—a frozen encoder with a light trainable head, not the official DOFA segmentation recipe.)

4.5. Supporting decomposition: what carries missing-band robustness, and when

Reliability under missing bands presupposes some accuracy-robustness to build on, and the operational question is which training ingredient supplies it. Holding the architecture fixed and varying only the training-time band-sampling rule, we compare band-group dropout against ChannelViT's hierarchical channel sampling [HCS, 3] and DiChaViT's diverse channel sampling [DCS, 13]. On CloudSEN12 (10 seeds) our dropout is statistically tied with HCS at $m = 0, 2, 3$ missing groups, shows a small significant advantage at $m = 1$ (+0.19 mIoU), and is beaten by HCS at $m =$ 4–5; the sampler's training missing-fraction coverage sets where on the degradation curve it is strongest, so "our dropout equals HCS" is false. DCS is significantly below HCS at intermediate missing (-0.48 to -0.69 mIoU, $m =$ 1–4, confidence intervals excluding zero): diversity sampling does not transfer to redundant spectral bands.

More consequentially, the missing-band robustness that band-group dropout supplies is itself regime-dependent. Under *synthetic* whole-group zeroing—the degradation that matches the augmentation—band-group dropout is the workhorse of missing-band robustness, as a companion decomposition on hyperspectral benchmarks establishes (supplementary material). Under *physical* degradation, however (6S gaseous transmittance with column water vapour, a band-scaling rather than band-zeroing corruption), that robustness does not transfer: the no-dropout MLP (b1) is uniformly *more* robust than the dropout MLP (b2) across the water-vapour grid, and indeed on the benchmark clean condition ($b2 - b1 = -8.2$ mIoU clean, -8.9 at CWV 2.0, -9.0 at CWV 4.0). Robustness measured under synthetic band removal therefore overstates operational robustness under physically-grounded band loss—the same synthetic-proxy caution that motivates measuring reliability under real, not simulated, shift.

4.6. A real-uncertainty anchor, and its limits

A natural objection is that the "unreliability" we measure could be an artifact of the labels or the model, with no independent physical referent. EMIT [8], an operational spaceborne imaging spectrometer, provides one: its L2A product ships a per-pixel, per-band posterior *retrieval* uncertainty from the ISOFIT inversion [17], grounded in the physics of the retrieval rather than in any label. On 14 EMIT granules spanning an NDVI range from bare desert to dense canopy, the alignment between our per-band reconstruction anomaly and this retrieval uncertainty is present but weak and mixed: the per-band rank correlation averages $+0.27$ (range -0.26 to $+0.69$), the per-pixel correlation only $+0.04$ (-0.52 to $+0.55$), with no clear trend against scene NDVI. We report this as an *observational* summary over 14 granules ($N = 14$ does not support a permutation test) and treat EMIT as a *supporting, not load-bearing* anchor: the retrieval uncertainty is itself a proxy, and the WorldCover labels [22] used to pair it with land cover are weak ($\sim$74 % global overall accuracy). The load-bearing evidence remains the CloudSEN12 conformal experiments.

5. Discussion and limitations

We lead with the limitations, because several bound the claims directly.

Coverage is part of the result, not a footnote.. The certified quantity is a joint mass, which any method drives to zero by abstaining. Mondrian's restoration is therefore genuine but partial: on the flagship it accepts 47 % of pixels where naive accepts 94 %. The operational reading is that per-stratum recalibration converts a silent, uncontrolled error rate into an *explicit* accuracy–coverage trade the operator can see and choose. We report every risk with its coverage precisely so this trade is never hidden.

Whether the crisis occurs is conditional.. The surface and geographic axes are both pure deployment shifts on a fixed input, so their contrast is clean: the structural surface gap breaks the naive certificate (11.4 %, CI excludes 10), the milder geographic gap does not (9.5 %, CI just below 10). The failure requires a structural domain gap between calibration and deployment, not mere geographic distance. The atmospheric shift breaks harder (28.9 %) but also changes the input representation, so we read it as a stronger, different-mechanism confirmation. We do not claim a smooth monotone "severity $= f(\text{gap size})$" law—an independently-measured gap metric across more shift types would be needed—but the surface-breaks / geography-holds contrast is a genuine diagnosis of *when*. The crisis is also strongly model-class-dependent (a foundation model is far more reliable), so it is not an artifact of a weak backbone.

Scope of the flagship evidence.. The atmospheric certificate is demonstrated on one benchmark (CloudSEN12) and one atmospheric processor (Sen2Cor); the CloudSEN12 test metadata records eight distinct Sen2Cor product-baseline versions, a mild internal heterogeneity we do not control for. A single-processor, single-benchmark flagship cannot establish that the *magnitude* generalises to every operational pipeline; the surface and geographic axes add evidence that the *phenomenon* is not pipeline-specific, on the same task and sensor. Extending to a second real atmospheric processor and to additional cloud benchmarks is the most direct strengthening.

The mechanism is borrowed; the comparison is not compute-matched.. Our band-as-modality architecture is a substrate of cited components, and the proposed-vs-MLP comparison is not matched on training compute, optimiser steps, or data exposure. The paper's claim is the *within-method* naive-vs-Mondrian gap, invariant to that confound because both arms share the trained model; the plain-MLP baseline reproducing the surface-axis result ($11.9 \rightarrow 2.5$ %) shows the finding does not depend on it.

Statistical caveats.. The exchangeable unit is the scene-connected component and the standard error is two-way cluster-robust; these were found by adversarial review and change the confidence intervals, not the direction of the result. Barber et al. [4]'s coverage-gap bound is worst-case and does not predict our exact numbers. The estimand is the mean risk on a fixed per-patch pixel sample (one sampling seed across runs), so our standard error reflects the calibration-split and model-seed variation, not pixel-sampling variability—a residual source we bound by using several hundred pixels per patch but do not fully propagate.

Implications.. For operational Earth observation the practical message is narrow and actionable: a model whose *accuracy* is robust to a shift—maintained or only moderately degraded—must not be assumed *reliable* under that shift, and a conformal or selective-prediction operating point calibrated before deployment should be treated as valid only within the calibration regime. Where the deployment regime is a known operational state, per-stratum recalibration is a cheap and principled safeguard whose only cost—abstention—is one the operator can see and choose. Where the shift is continuous or unknown, the weighted- and adaptive-conformal machinery we deliberately did not invoke is the natural next tool, and quantifying its cost under the same operational shifts is future work.

6. Conclusion

Robustness certification is not reliability certification. On real Sentinel-2 cloud segmentation, under the operational L1C→L2A atmospheric-correction shift a model's accuracy degrades only moderately (a 12-point drop) while a naive conformal certificate comes to admit more than three times the confidently-wrong risk it reports—silently, because the statistic the certificate minimises still passes. The failure is not confined to that one shift, nor is it universal: two pure deployment shifts pin down the condition—a structural surface gap breaks the naive certificate, while a milder geographic gap does not—so the reliability crisis requires a structural domain gap rather than any shift. A channel-adaptive foundation model is far more reliable, so the severity is also model-class-dependent. Per-stratum recalibration restores control wherever the certificate breaks, at an abstention cost we report rather than hide, and a defensible certificate on this data requires a scene-connected-component exchangeable unit and a cluster-robust variance that naive alternatives get wrong. We contribute what is, to our knowledge, the first operational remote-sensing instantiation of the robustness $\neq$ reliability decoupling together with a diagnosis of when it occurs—a small, honest, and we hope useful correction to how spectral segmentation models are trusted in deployment.

Data and code availability

Code and committed result tables are available in the project repository. CloudSEN12 [2], EMIT and ESA WorldCover are public.

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